

INTERANNUAL VARIABILITY IN GROWTH RATES OF EARLY JUVENILE SABLEFISH AND THE ROLE OF ENVIRONMENTAL FACTORS

Susan M Sogard

ABSTRACT

Otolith increments were examined to estimate growth rates of late larvae and early juvenile sablefish from an 8-yr time series (1993–2001), with fish collected from late March to mid-May at the same location off Oregon. The time series spanned a broad range of oceanographic conditions and included an extreme El Niño year (1997–1998) followed by an extreme La Niña (1999). Estimated growth rates overall were rapid for the small size of fish examined and the water temperatures they experienced, averaging 0.4 mm d^{-1} to 1.0 mm d^{-1} at fish sizes of 10–40 mm. Growth rates were strongly related to fish size and water temperature. After adjusting for these effects using regression residuals, interannual differences in growth corresponded to large-scale oceanographic indices of productivity and transport in the California Current ecosystem. Growth during the early neustonic stage was correlated with subsequent recruitment to the adult stock except in 1997, when conditions in late winter and early spring apparently were favorable for sablefish growth but survival of the year class was poor, potentially a consequence of reduced productivity as El Niño conditions developed in the summer. Within a year, there was minimal support for the occurrence of growth-dependent mortality, at least within the time frame of our collections. Overall, the results confirm the unusual capacity for rapid growth in small sablefish and are consistent with the general pattern of large-scale environmental controls on larval and early juvenile survival hypothesized for fishes resident in the California Current.

Variability in environmental conditions during early life stages can have dramatic effects on survival and subsequent recruitment to the adult stage. For many marine teleosts, the abundance of annual cohorts (year class strength) is presumed to be determined during larval or early juvenile stages (Houde 1987). Thus, understanding environmental factors that modify early growth and survival rates is vital to predicting favorable and unfavorable conditions for population persistence and resilience.

Sablefish [*Anoplopoma fimbria* (Pallas, 1814)] are long-lived (> 100 yrs, Beamish and McFarlane 2000) and broadly distributed, with adults residing in outer shelf and slope habitats from California throughout the northern Pacific Ocean to Kamchatka. In contrast to the slow-growing adults, which reside at depths with cold temperatures, presumably stable but limited food availability, and often low oxygen levels, larvae and early juveniles reside in surface waters, where they encounter warmer temperatures and variable food availability dependent on local productivity conditions. Spawning occurs in winter months, with eggs released and fertilized at depths of 100s to 1000s of meters (Mason et al. 1983). Incubation time is dependent on temperature, but averages around 2 wks at 5.5°C (Alderdice et al. 1988). After hatching at a size of 5–6 mm SL (Kendall and Matarese 1987), the yolk sac larvae gradually migrate to surface waters (McFarlane and Beamish 1992). During this migration they

begin to feed on early stages of calanoid copepods. After full yolk sac absorption, the larvae are neustonic, with limited movement downward in the surface layer during the day and restricted occurrence in the neuston at night (Sogard and Olla 1998). Transformation from the larval to juvenile stage is a gradual process lacking definitive benchmarks, but Kendall and Matarese (1987) concluded that larval development is complete at a size of 35 mm SL. Juveniles remain in surface waters for 1–2 yrs depending on the population, then migrate to their adult benthic habitats.

Young-of-the-year (YOY) sablefish exhibit a remarkable capacity for rapid growth. In laboratory experiments, they have attained growth rates of up to 3 mm d⁻¹, with a clear dependence of growth on temperature (Sogard and Olla 2001). Field estimates of growth based on otolith increments have suggested rates averaging 1.19 mm d⁻¹ for large (70–200 mm) YOY fish in an Alaskan population (Sigler et al. 2001) and 1.47 mm d⁻¹ for fish 103–260 mm in a West Coast population (Boehlert and Yoklavich 1985). In contrast to many teleosts, juvenile sablefish have a limited capacity for compensatory growth and appear to maximize consumption under all conditions (Sogard and Olla 2002), suggesting a highly risk-prone foraging strategy and intense selection pressure for rapid growth during their residence in the neuston layer.

Two populations of sablefish are currently recognized: an Alaskan population ranging from north Vancouver Island through the Aleutian Islands and a West Coast population ranging from south Vancouver Island to southern California (Kimura et al. 1998). For the West Coast population, growth opportunities for neustonic larval and early juvenile stages are subject to the highly variable surface environment of the California Current. Fluctuations in the primary factors of temperature and the quantity and quality of available zooplankton prey have been extensively studied in this system. It is now well recognized that the California Current has multiple atmospheric forces that drive circulation and production on multiple temporal and spatial scales. The Pacific Decadal Oscillation (PDO) index provides a basin level metric of two major phases or regimes that affect the California Current ecosystem (Mantua et al. 1997). During the positive or warm regime, coastal surface temperatures are higher, upwelling and thereby nutrient input to surface waters is lower, and overall primary productivity is reduced compared to the negative or cool regime. These contrasting environmental conditions are thought to have major impacts on lower trophic levels, with clear bottom-up effects on fish populations (Francis et al. 1998, Peterson and Schwing 2003). The El Niño–Southern Oscillation (ENSO) has comparable broad spatial scale effects on conditions supporting productivity in the California Current, but operates on a shorter 3–7 yr time scale (Lehodey et al. 2006). Localized variability in circulation patterns and upwelling can exacerbate or moderate the primary physical environment set up by the ocean conditions indexed by PDO and ENSO.

Recent stock assessments have indicated high variability in sablefish recruitment over the last 40 yrs, and physical patterns in the ecosystem appear to underlie much of this variability (King et al. 2000, 2001, Schirripa and Colbert 2006). For the West Coast population, Schirripa and Colbert (2006) found significant roles for three oceanographic indices in correlations with recruitment indices. The north-south component of Ekman transport in February at 48°N latitude provides an index of the shift from poleward (positive values) to equatorward flow (negative values) along the West Coast; higher sablefish recruitment levels were observed when February transport had already shifted to an increased southerly component. This index

reflects conditions during egg incubation, hatching, and development of yolk sac stages. An index of east-west Ekman transport in June, during a period of high growth potential for early juvenile stages, was also correlated with sablefish recruitment. More westward Ekman transport, an indicator of greater upwelling, was associated with relatively high recruitment from 1967 to 1980, whereas more eastward transport from 1981 to 2002 was associated with generally low recruitment. Mid-coast sea level in July was also negatively correlated with recruitment, suggesting improved survival of juveniles during higher southward flow and greater upwelling (as indexed by lower sea level). McFarlane and Beamish (1992) concluded that year class strength in sablefish was related to very early larval survival, in turn dependent on copepod production for post yolk sac stages during their upward migration to surface waters. However, the importance of climate factors in late spring and early summer (Schirripa and Colbert 2006) suggests that events later in the neustonic stage may also play a role in fine-tuning survival.

A potential mechanism affecting larval/juvenile sablefish survival under contrasting oceanographic conditions is the quality of available prey and consequent effects on growth rates. In the California Current, shifts in large-scale flow patterns induce concomitant shifts in the zooplankton community. With more flow from the south, copepod diversity is higher and the community is dominated by smaller, subtropical species, whereas copepod diversity is lower and dominated by larger, boreal species when flow increases from the north (Hooft and Peterson 2006). Grover and Olla (1987) documented a shift in juvenile sablefish diets during an El Niño year to a smaller species of copepod common to the southern zooplankton community, in contrast to a "normal" year, when sablefish primarily consumed a larger species common to the northern community. This shift in diet is thought to have nutritional consequences for early life history stages of fishes. Lee et al. (2006) found that northern zooplankton species in general store more lipid compared to southern species, suggesting that they provide more energy for the larval fishes that consume them. In addition to the physical environmental factors found to be correlated with sablefish recruitment, Schirripa and Colbert (2006) also noted a strong, positive relationship of recruitment with a greater dominance of the copepod community by boreal species.

In the present study, otolith microincrements were used to examine growth rates of larval and early juvenile stages of sablefish from a series of collections off the coast of Oregon from 1993 to 2001. It was hypothesized that bottom-up processes influencing prey quality and quantity determine growth rates of early stages of sablefish, with consequent effects on survival and ultimately recruitment. In the California Current ecosystem, atmospheric processes such as wind speed and direction are primary drivers of large-scale variability in ecosystem productivity (Wells et al. 2008), and can be expected to impact the growth environment for small consumers such as juvenile fishes. The objectives of the present study were to: (1) estimate natural growth rates for comparison with growth determined under laboratory conditions; (2) test for the occurrence of growth-dependent mortality within discrete intra-annual cohorts; (3) examine the role of the oceanographic environment on growth; and (4) determine correlations of growth during the early neustonic stage with subsequent recruitment or year class strength.

METHODS

SAMPLE COLLECTION AND OTOLITH PREPARATION.—Sablefish were collected on chartered commercial fishing vessels using two nets identical to those described by Shenker (1988). The manta net was 1 m wide by 0.7 m deep and had a mesh size of 1 mm. The neuston trawl was 3.5 m wide by 1.05 m deep and had a mesh size of 4.8 mm. Nets were deployed off of the side of the vessel using outriggers and skimmed the water surface with no rigging in the water ahead of the net mouth. All tows were conducted at night to reduce net avoidance by fish and were 10 min in duration with a boat speed of ~1.5 m s⁻¹. Flowmeters deployed in the center of the nets on seven of the cruises confirmed an average surface area of 1 km² sampled by the manta net and 3.5 km² for the neuston trawl in a 10 min tow. Samples were sorted on deck and all sablefish were either returned to the laboratory alive for behavioral studies or preserved in 95% ETOH for subsequent otolith analysis.

All sablefish were collected within the same general region, ~40 km offshore of Newport, Oregon, near the shelf/slope break, with bottom depths of 50–300 m. Of 1289 tows (all years combined), 94% were within a 15 km radius of a mean location at 44.78°N and 124.56°W. Fishing was conducted from 1993 to 2001, with three trips each year from March to May (Table 1). Not all trips produced sufficient numbers of fish for otolith analysis; in 1996, too few samples were available for any of the trips. The primary objective of sampling was the collection of live fish rather than a survey of abundance. Consequently, it was not possible to derive reliable estimates of density from these tows.

Lapillar otoliths were dissected from individual fish, mounted on microscope slides with Crystal Bond adhesive, and polished in a frontal plane on both sides. Daily deposition of sagittal increments was noted by Boehlert and Yoklavich (1985) and preliminary laboratory experiments also demonstrated daily deposition in lapilli (Sogard, unpubl data). Increments were counted and widths between successive increments were measured along a common axis from the core to the middle of the medial side using Optimas software. Paired lapilli from 71 individuals were analyzed to determine potential differences between right and left otolith. Both increment counts and otolith radii measurements were closely matched (coefficient of variation < 5%), allowing either otolith to be used for overall analyses. For individuals in which both lapilli were measured, mean values were used in all data analyses.

OCEANOGRAPHIC INDICES.—Sea surface temperatures (SST) were estimated for each date during the neustonic residence of all fish analyzed. Temperatures within the core collection region were derived from the Integrated Global Ocean Services System (IGOSS, <http://iridl.ldeo.columbia.edu/SOURCES/IGOSS>) using a location of 44.5°N, 124.5°W. For the date of each collection, the IGOSS temperatures closely matched those measured directly from the boat, suggesting accurate estimates for the capture location. Temperatures experienced by individual fish on each day prior to capture were estimated by matching to the weekly mean SST.

Table 1. Date and sample size (n) of neuston tows off Oregon used for analysis of sablefish otoliths. In 1996, too few fish were collected for analysis.

Year	Cruise 1	Cruise 2	Cruise 3	Number fish analyzed
1993	Apr 4	Apr 19	May 3	32, 11, 21
1994	Mar 30	Apr 17	Apr 28	11, 9, 13
1995	Apr 5	Apr 18	May 9	8, 16, 6
1997		May 1	May 19	15, 8
1998	Apr 2	Apr 17		14, 11
1999	Mar 31	Apr 29	May 18	5, 27, 31
2000		May 7		20
2001	Apr 23	May 16		21, 16

Monthly values of the Pacific Decadal Oscillation (PDO) were obtained from the Joint Institute for the Study of the Atmosphere and Ocean (JISAO, <http://jisao.washington.edu>). Bi-weekly values of sea level height were obtained from the University of Hawaii Sea Level Center (<http://ilikai.soest.hawaii.edu/uhs/c/html/0592A.html>). The site location of South Beach, Oregon, was used as the nearest location for our collections. Monthly values of the Northern Oscillation Index (NOI) were obtained from the NOAA/NMFS Environmental Research Division (ERD) Live Access Server (<http://www.pfeg.noaa.gov/products/las.html>). Monthly indices of offshore (east-west) Ekman transport (upwelling) and alongshore (north-south) transport at 44.5°N, 124.5°W were also obtained from the ERD website. These indices are intercorrelated to varying degrees, but are derived from different metrics. In general, oceanographic conditions indicating reduced upwelling and reduced equatorward transport were expected to reflect reduced primary and, consequently, secondary production as well as reduced transport of lipid-rich boreal copepods into the study area.

DATA ANALYSIS.—All growth estimates were limited to the period after fish were presumed to have completed migration to surface waters. 8 mm total length (TL) was used as a cutoff of full recruitment to the neuston, based on size distributions in our manta net collections over the time series. Recent growth was estimated by the mean width of the five outermost lapillar increments, excluding the final increment. These values provided an index of growth during the week prior to collection. Overall growth during the early neuston period was estimated by the full series of increments for each individual, excluding the final increment and all increments deposited prior to the fish attaining an estimated size of 8 mm TL; i.e., an otolith radius of 18.62 μ as determined by the otolith radius to fish TL relationship. The final increment was excluded because it might represent less than a full day's growth. Exclusion of the earliest increments ensured that growth estimates began after fish had arrived in the neuston.

Growth-dependent mortality was evaluated by tracking cohorts within each year of fish collection. Individuals within a cohort were assumed to experience a common environment during the period of increment deposition. Cohorts were defined as subsets of individuals that were the same size at the same time within a 20-d interval, based on otolith radius measurements at each increment. Thus, all fish that obtained a size of 10 mm TL (corresponding to a lapillar radius of 20.53 μ) within a 20-d interval were considered a cohort. Date at each increment was determined by counting increments backwards from the margin and subtracting from the date of capture. Designation of cohorts was based on sample sizes of fish with similar temporal availability within each year; individuals not attaining 10 mm within the 20-d interval were excluded from the analysis. Thus, cohorts were defined based on the sampling schedule and do not reflect temporal pulses in recruitment. Over the full time series, 10 discrete cohorts (two in 1993 and 1994, one in all other years) were defined. Two sets of increment widths were compared. The first was calculated as the mean increment width during the earliest size interval of neuston residence, when fish were estimated to be between 8 and 12 mm TL based on otolith radius (18.62–35.05 μ) at each increment and the otolith size to fish size relation. The second growth interval was the mean increment width, when fish were estimated to be between 15 and 20 mm TL (45.62–62.52 μ radius). For each cohort, mean increment widths during each size interval were regressed against fish length. If growth-dependent mortality occurred, larger fish (survivors) should have had either wider increments compared to smaller fish, suggesting selection for faster growth, or narrower increments, suggesting selection for slower growth, during an earlier growth interval. ANCOVAs were also performed for all cohorts pooled with length as a covariate.

To examine interannual differences in growth and their potential correlation with recruitment indices, a growth index was derived for each fish that adjusted increment widths for influences due to the size of the fish and the water temperature on the day of increment deposition. The widths of all increments deposited after fish had attained an estimated size of 8 mm TL (and therefore were presumed to be residing in surface waters) were run in a multiple regression that included daily temperature estimates and daily otolith radius estimates. The regression was highly significant for both variables, with growth increasing with size at incre-

ment deposition and with warmer temperatures (see Results). Residuals for each increment were calculated from the multiple regression and the mean residual for each fish was presumed to provide an index of its overall growth relative to other fish during its early neustonic residence. Positive mean residuals indicated that a fish had higher than expected growth relative to its body size and the temperatures it experienced. Negative residuals indicated that fish grew relatively slowly for their individual circumstances. Mean growth indices by year were compared with ANOVA. Growth indices were also compared with environmental variables and the recruitment indices for each year as estimated by the most recent assessment for the West Coast sablefish stock.

RESULTS

The otolith size to fish size relationship was isometric within the size range (14–51 mm TL) of sablefish used in this study ($r = 0.96$, $P < 0.001$, $n = 295$; Fig. 1). Fitting a series of polynomial models did not improve the correlation (all $r \leq 0.96$), suggesting that the fish size to otolith size relationship remained proportional during the limited size range examined here. In addition, a high positive correlation ($r = 0.74$) of the residuals of a regression of TL on increment counts with the residuals of lapillar radius on increment counts indicated minimal age-independent variability in the proportionality of otolith size to fish size (Hare and Cowen 1995). Growth effects on the relationship, whereby faster growing fish have relatively small otoliths (Campana 1990), were minimal based on the low correspondence of the residuals of increment counts on fish size with the residuals of lapillar radius on fish size ($r = 0.45$, Hare and Cowen 1995). Residuals of all regression relationships were normally distributed based on Kolmogorov-Smirnov tests (all $P > 0.05$). In addition, the statistical intercept (5.11) of the regression of TL on lapillar radius was within the estimated size range at hatching (5–6 mm, Kendall and Matarese 1987), again suggesting that growth effects were minimal. It is not known when the lapillus is formed in larval sablefish or when the first increment is deposited. Because precise values of potential biological intercepts of fish and otolith size at hatch were not known, growth

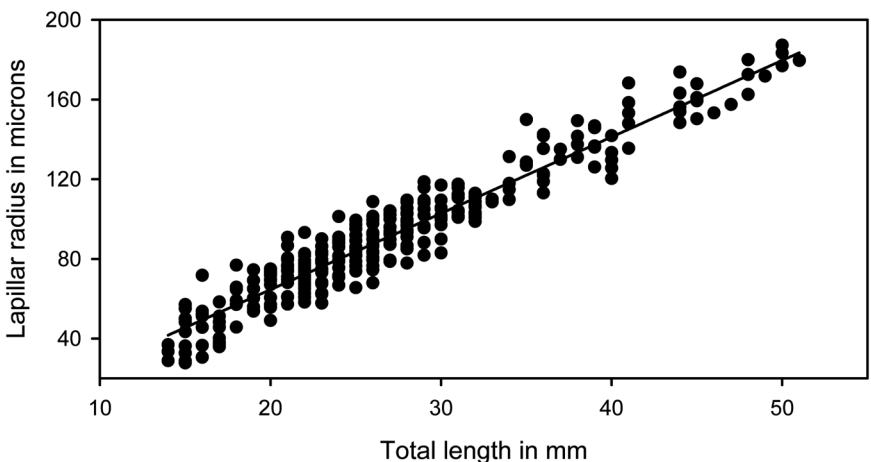


Figure 1. Regression of lapillar radius (in μ) on total length (in mm) for 295 larval/juvenile sablefish collected from 1993 to 2001 offshore of Newport, Oregon. Line is $P < 0.001$, $r = 0.96$.

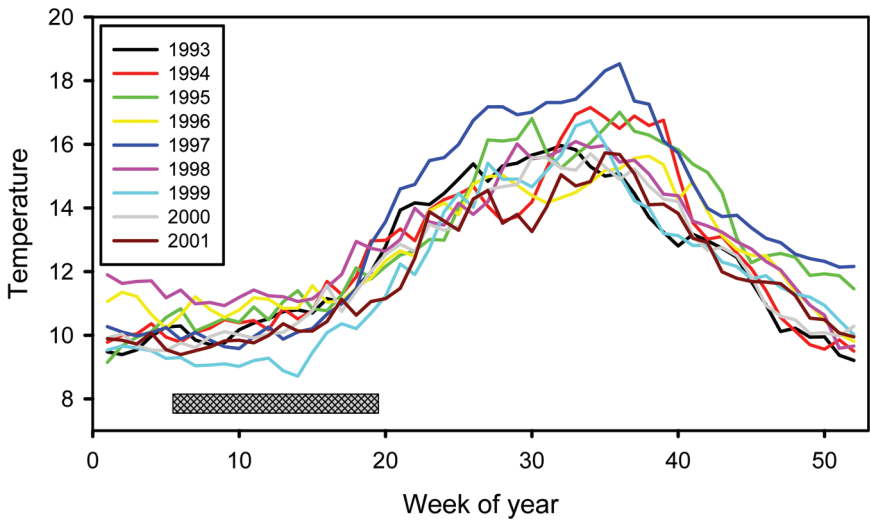


Figure 2. Mean weekly sea surface temperature from 1993 through 2001 at 45.5°N, 124.5°W, the approximate midpoint of sablefish sampling. Hatched bar indicates the time period of otolith increment deposition for fish collected during this study.

estimates were directly estimated from the regression calculated on fish at capture. Based on this relationship, fish that were 8 mm TL had an otolith radius of $\sim 18.62 \mu$. Assuming that all fish ≥ 8 mm had completed migration to surface waters, increment counts back to this radius suggested that juvenile sablefish collected in this study were present in the neuston layer beginning in early February.

Temperatures experienced by juvenile sablefish during the early neustonic period examined in this study ranged from 8.7 to 13.6 °C and varied by year (Fig. 2). The time series of our collections included one of the most intense El Niños recorded along the West Coast. Water temperatures began increasing rapidly in late April of 1997 and remained higher than average throughout the following summer, fall, and winter months. The El Niño was followed by an intense La Niña, clearly evident in low temperatures during the winter and spring of 1999. As a consequence, the 1998 sablefish year class was exposed to temperatures about 2 °C higher than those of the 1999 year class during their initial neuston residence. However, it was the 1997 year class that experienced the highest temperatures during the summer and fall seasons.

Recent growth (five outermost increments) was highly dependent on fish size and water temperature (Fig. 3, Table 2). A multiple regression including these two factors indicated that a large proportion of the variation in mean increment width was related to positive effects of both factors ($P < 0.001$ for each factor, model $r^2 = 0.584$, $n = 295$). As body size increased, the capacity for growth apparently increased dramatically. For example, at a given temperature, fish 50 mm in length on average had outer increments that were 2.1 μ wider than those of fish 15 mm in length. At a given size, fish in 12 °C water had outer increments that were 0.6 μ wider than those of fish residing in 10 °C water. Based on the linear correspondence of otolith size with fish size, these differences suggested an estimated increase in daily growth of 0.015 mm for each mm increase in total length, and 0.08 mm for each 1 °C increase in SST.

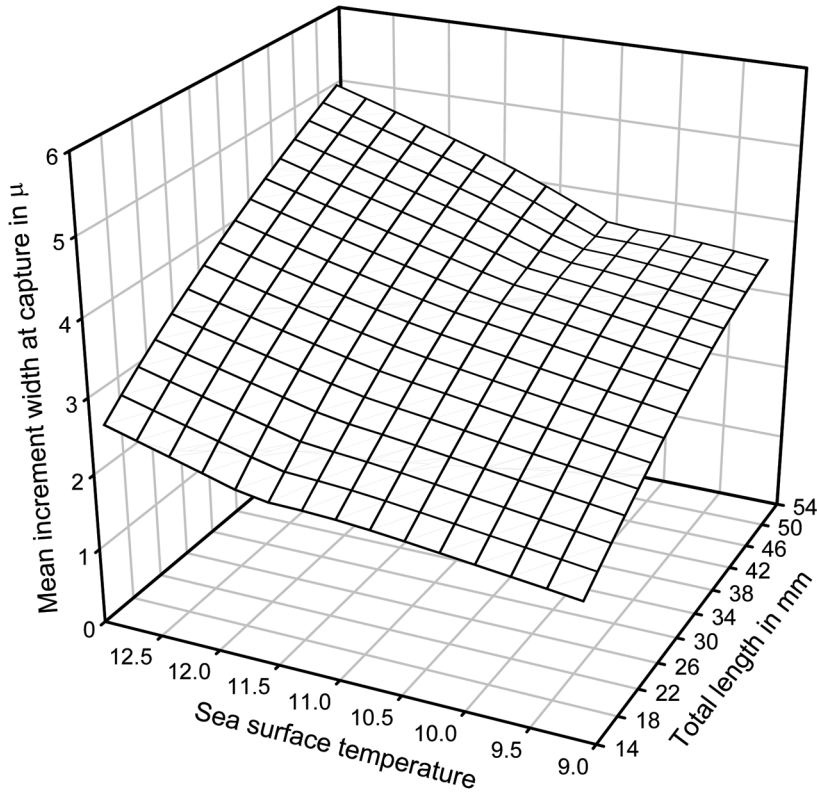


Figure 3. Mean widths of otolith increments deposited at capture (outer five) for 295 larval/juvenile sablefish vs total length and water temperature at time of capture. Mesh plot was derived from loess-smoothed values for each individual. All years (1993–2001) combined.

Because as fish aged they also typically experienced warmer temperatures (Fig. 2), the two factors of body size and temperature acted in concert to increase fish growth rates with time. For the temperatures observed during the neustonic period of the fish collected over the time series, estimated growth rates averaged around 0.4 mm d⁻¹ for fish at 10 mm TL and 1.0 mm d⁻¹ when fish had reached a size of 40 mm TL. Based on laboratory length-weight relationships, these absolute growth rates in length are comparable to specific growth in weight of ~2.6% d⁻¹ at a size of 10 mm TL and 6.6% d⁻¹ at a size of 40 mm TL.

Table 2. Results of multiple regression of otolith increment widths at capture (outer five increments excluding last increment) on total length and water temperature (n = 295 fish).

Factor	Regression coefficient	Standard error	Standardized regression coefficient	F	P	r ²
Constant	-1.974	0.643	0.000	9.42	0.002	
Total length	0.059	0.004	0.621	188.18	< 0.001	
Temperature	0.310	0.064	0.219	23.36	< 0.001	
Model				204.95	< 0.001	0.584

Table 3. Intra-annual sablefish cohorts examined for potential evidence of growth-dependent mortality during the early neustonic period (fish size 8–12 mm TL). Date = mid-date of the 20-d time period in which fish attained an estimated size of 10 mm TL. N = number of fish in the cohort. TL = size range in mm. Correlation statistics are based on standard linear regression of TL on increment width at size interval 8–12 mm.

Year	Date	n	TL in mm (range)	Mean increment width in μ (SE)	Standardized regression coefficient	P
1993	Mar 10	40	19–45	1.55 (0.05)	–0.16	0.323
1993	Mar 30	22	16–46	1.56 (0.06)	–0.32	0.151
1994	Mar 2	12	15–32	1.47 (0.06)	0.17	0.603
1994	Mar 25	21	18–34	1.44 (0.06)	–0.42	0.06
1995	Mar 30	29	15–40	1.73 (0.08)	0.22	0.242
1997	Apr 9	12	15–48	1.78 (0.10)	0.07	0.826
1998	Mar 20	17	15–32	1.49 (0.04)	0.28	0.269
1999	Apr 9	39	16–41	1.57 (0.06)	0.56	< 0.001
2000	Mar 29	18	28–51	2.11 (0.15)	0.04	0.874
2001	Apr 8	23	14–36	1.58 (0.05)	–0.22	0.313

Early growth rates of larger fish (survivors) in discrete cohorts did not provide strong evidence of growth-dependent mortality. For the earliest period of neuston residence, when fish were 8–12 mm TL, only one cohort (1999) had significantly higher growth rates in larger fish compared to smaller fish [$P < 0.001$, Table 3, an ANCOVA of all cohorts pooled also indicated a non-significant effect of TL as a covariate ($P = 0.507$)]. During the size interval of 15–20 mm TL, the first 1993 cohort had a significant relationship between size and growth ($P = 0.026$), but it was negative, with faster growth for smaller sizes of fish. None of the other cohorts had a significant relationship of growth with body size [all $P > 0.05$; Table 4, an ANCOVA of all cohorts pooled also indicated a non-significant effect of TL as a covariate ($P = 0.441$)].

After adjusting otolith increment widths for otolith size (a proxy of fish size) and SST on the day of deposition, the derived growth indices revealed significant differences by year (ANOVA: $F_{7,287} = 6.32$, $P < 0.001$). Two general groups were evident,

Table 4. Intra-annual sablefish cohorts examined for potential evidence of growth-dependent mortality during the neustonic period when fish were 15–20 mm TL. Date = mid-date of the 20-d time period in which fish attained an estimated size of 10 mm TL. N = number of fish in the cohort. TL = size range in mm. Correlation statistics are based on standard linear regression of TL on increment width at size interval 15–20 mm.

Year	Date	n	TL in mm (range)	Mean increment width in μ (SE)	Standardized regression coefficient	P
1993	Mar 10	40	19–45	2.42 (0.06)	–0.35	0.026
1993	Mar 30	21	16–46	2.52 (0.07)	0.22	0.333
1994	Mar 2	11	21–32	2.26 (0.07)	0.18	0.596
1994	Mar 25	21	18–34	2.64 (0.06)	–0.41	0.067
1995	Mar 30	22	19–40	2.49 (0.07)	0.06	0.775
1997	Apr 9	11	18–48	2.86 (0.09)	–0.24	0.480
1998	Mar 20	13	18–32	2.59 (0.11)	0.43	0.139
1999	Apr 9	39	16–41	2.61 (0.04)	0.14	0.409
2000	Mar 29	18	28–51	2.81 (0.08)	–0.29	0.248
2001	Apr 8	17	16–36	2.45 (0.06)	0.20	0.395

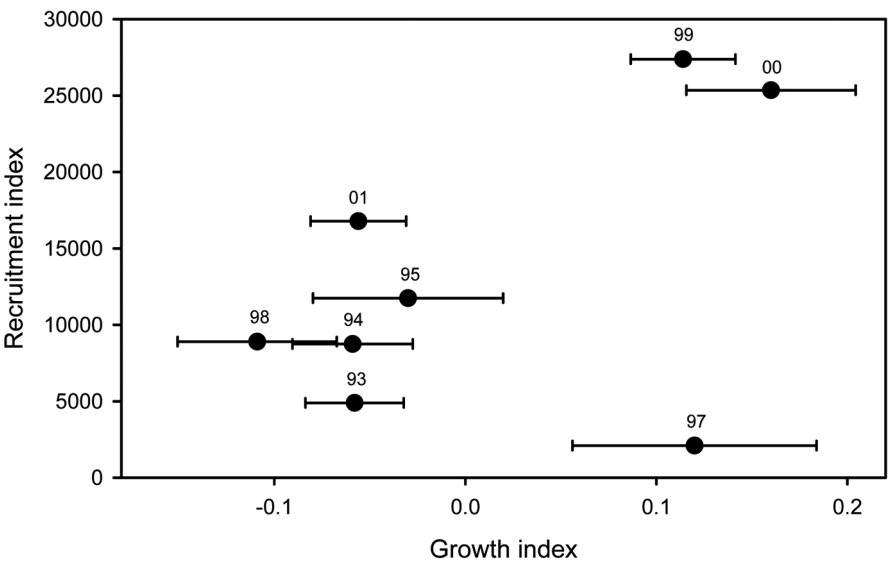


Figure 4. Recruitment index (number of age-0 fish) from the sablefish stock assessment vs growth index for early neustonic stage sablefish in years 1993–2001 off Newport, Oregon. Growth index was based on the mean residual of individual fish after regressing increment widths on radius at increment and sea surface temperature at increment. Numbers above data points designate year.

with relatively fast growth in 1997, 1999, and 2000, and relatively poor growth in 1993, 1994, 1995, 1998, and 2001 (Fig. 4). Monthly indices of oceanographic conditions were evaluated in stepwise multiple regression to determine potential correlations with sablefish growth over the 8 yrs of the time series. Several indices were relatively highly correlated with the growth index, the highest of which was the February PDO ($r = -0.825$). Due to intercorrelations of the various indices, only one additional variable, March alongshore transport, was included in the regression model following adjustment for February PDO. The resulting model suggested that these two factors accounted for a significant portion of the interannual variance in growth ($r^2 = 0.86$, $P = 0.007$; Table 5). Growth was higher when the February PDO was negative and when northerly Ekman transport in March was low.

There was a close correspondence between the growth index for each year and estimated recruitment based on the West Coast sablefish stock assessment, with the

Table 5. Results of stepwise multiple regression comparing the growth index of juvenile sablefish over an 8-yr time series with environmental indices from January to April for each year. Only factors included in the final model are listed.

Factor	Regression coefficient	Standard error	Standardized	F	P	r ²
			regression coefficient			
Constant	−0.007	0.023	0.000	0.09	0.775	
February PDO	−0.069	0.024	−0.561	8.13	0.036	
March northern Ekman transport	−0.000	0.000	−0.499	6.43	0.052	
Model				15.38	0.007	0.86

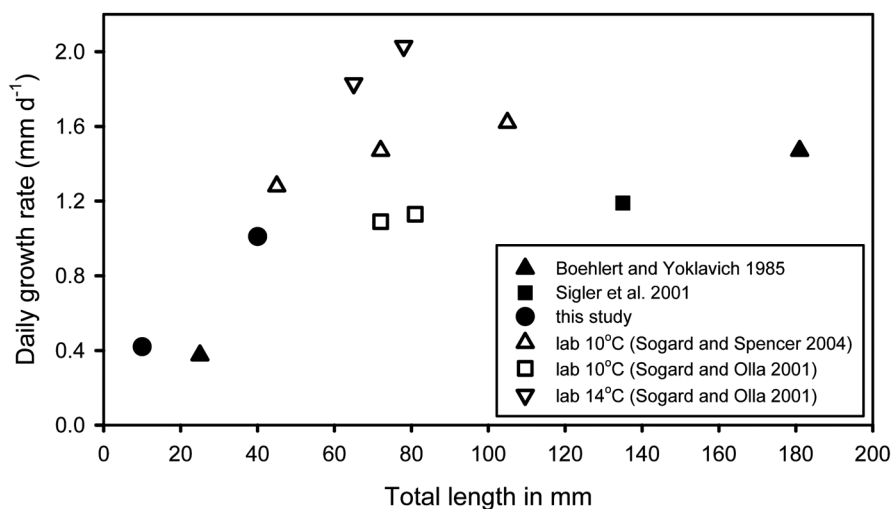


Figure 5. Daily growth rates of juvenile sablefish estimated in the field off Oregon and Washington for fish in two size classes (10–41 mm and 103–260 mm TL, Boehlert and Yoklavich 1985), for fish in an Alaska population at 70–200 mm TL (Sigler et al. 2001), and in this study for fish at an average size of 10 and 40 mm TL; and in lab experiments for fish maintained on ad libitum rations at 10 or 14 °C (Sogard and Olla 2001, Sogard and Spencer 2004).

exception of 1997 (Fig. 4). If 1997 is excluded, the correlation for the other 7 yrs was highly significant ($r^2 = 0.79$, $P = 0.007$).

DISCUSSION

Over the range of temperatures (9–14 °C) and oceanographic conditions of the 8 yrs of this study, estimated growth rates averaged around 0.4 mm d⁻¹ for 10 mm sablefish and 1.0 mm d⁻¹ for 40 mm TL fish. Growth increased dramatically as body size increased and water temperatures became warmer. The average growth rates estimated in this study are comparable to those calculated from sagittal increment counts by Boehlert and Yoklavich (1985) for sablefish collected in May off the coast of Oregon and Washington, presumably in water temperatures similar to those observed in this study. They estimated growth rates of 0.375 mm d⁻¹ for fish ranging from 10 to 41 mm SL and 1.47 mm d⁻¹ for fish ranging from 103 to 260 mm SL (Fig. 5). Growth rates estimated from otolith increments for an Alaskan sablefish population were slightly lower, at 1.19 mm d⁻¹ for fish 70–200 mm (Sigler et al. 2001). Prior studies of growth rates for fish held on ad libitum rations in the laboratory have used larger fish than those examined in the present study, but found comparable or slightly higher growth rates at a temperature of 10 °C (Fig. 5; Sogard and Olla 2001, Sogard and Spencer 2004). Raising water temperature to 14 °C resulted in much faster growth in the laboratory. Although the sizes of fish examined in the present study are not directly comparable to lab results, the general patterns observed suggest that early larvae/juveniles in surface waters in the late winter/early spring can grow at rates approaching those observed under ad libitum rations in the laboratory. These growth rates are extremely rapid for the small size of fish and moderate water

temperatures occurring in northeast Pacific waters during late winter/early spring. Faster growth rates have been recorded at similar sizes for other teleosts, but only for tropical species residing in much warmer temperatures. Yellowfin tuna larvae, for example, had estimated growth rates of $0.46\text{--}2.06\text{ mm d}^{-1}$ for fish 6–20 mm SL in temperatures of $26\text{--}30\text{ }^{\circ}\text{C}$ (Wexler et al. 2007).

Although the unusual growth capacity of early stages suggests strong selection in sablefish for rapid growth, cohort tracking provided only limited evidence of growth-dependent selective mortality. In the 1999 cohort, strong selection for faster growth was apparent during the 8–12 mm size interval, but not in the subsequent 15–20 mm interval. In the first 1993 cohort, no selection was evident in the 8–12 mm interval, but selection for slower growth during the 15–20 mm interval was suggested by a negative relationship between the growth index and fish size. Selective mortality was not evident in any of the other discrete cohorts. Samples of larger fish were insufficient to test for selective mortality in later size classes. With the expanding capacity for growth at larger sizes and warmer temperatures, the potential for greater variability in growth increases and may lead to greater selectivity. The intensity and direction of size or growth-dependent mortality can vary with ontogeny and among local populations as a result of smaller scale ecological attributes such as density (Johnson and Hixon 2010). Because our sampling focused on the collection of live fish for behavioral experiments and was not a standardized survey, it was not possible to derive reliable estimates of fish density. At the scale of year classes in this study, the correlation of growth with subsequent recruitment is consistent with expected higher mortality of slower growing year classes. In contrast to many teleosts, which have optimal growth rates that are lower than physiologically possible (Arendt 1997), early stages of sablefish appear to consistently grow at maximal rates for the environment experienced by the fish, suggesting that the benefits of rapid growth disproportionately outweigh the costs (Sogard and Olla 2002).

The growth estimates derived in the present study for the early larval/juvenile period (February–May) of sablefish at a single location off Oregon appeared to closely match broad-scale patterns of productivity in the California Current ecosystem. Growth improved when the PDO in February was negative. The PDO provides a large-scale correlate of productivity in the North Pacific (Mantua et al. 1997), with generally enhanced primary production in the California Current during the negative or cool phase. Favorable conditions for growth and survival associated with increased upwelling and other physical parameters generally characteristic of a negative PDO in the California Current have been reported for a broad ranges of species, including zooplankton, fish, and marine birds (Mantua et al. 1997, Brodeur et al. 2008, Wells et al. 2008). Although the PDO is often used to define multi-year cycles of favorable vs unfavorable productivity regimes, the monthly value potentially provides an index of finer scales of temperature conditions, in turn reflecting processes associated with bottom-up production supporting larval fish growth. Sablefish growth also improved when alongshore northerly transport in March was reduced. For their much longer time series, Schirripa and Colbert (2006) found a similar relationship, with higher sablefish recruitment in years with more southerly transport in February. Although linkages between oceanographic conditions and fish populations in the California Current have often emphasized the period following the spring transition, when primary productivity in surface waters generally tracks the supply of nutrients delivered by upwelling, oceanographic indices in late winter may also reflect overall

favorable conditions for larval fish growth and survival. Schroeder et al. (2009) provide a compelling argument that physical conditions in winter have major effects on ecosystem-wide productivity. Like sablefish, many California Current fishes spawn in winter months well before the spring transition (Parrish et al. 1981, Shanks and Eckert 2005, Brodeur et al. 2008) and their annual success may depend on productivity prior to as well as after initiation of the typical upwelling season.

As previously noted in lab studies, sablefish growth in the wild was closely correlated with temperature. Warmer temperatures provide a highly effective physiological mechanism enabling faster growth (Sogard and Olla 2001). However, productive oceanographic conditions in the eastern North Pacific, in general, entail relatively colder SST as a consequence of increased upwelling and enhanced equatorward transport. These conditions are presumed to both enhance primary productivity and transport more energy-rich boreal copepods to the northern California Current ecosystem (Hoof and Peterson 2006). Thus, there is an apparent contradiction in that warmer temperatures enhance the physiological capacity for growth in juvenile sablefish, but warmer temperatures in the environment are typically correlated with conditions of lower productivity. Consequently, the cooler temperatures sablefish would experience during negative PDO regimes and increased southward transport are likely offset by improved foraging success. For example, 1999 was the coldest year in the time series, but absolute growth, as estimated by increment widths, was intermediate relative to other years. After adjusting for temperature and fish size, the growth index was high, indicating that fish were growing better than expected. The negative correlation of sablefish growth with the PDO index is clearly not a function of warmer temperatures impeding growth; it more likely reflects the absence of sufficient energy resources to support rapid growth as the PDO becomes more positive.

There was a good correlation of annual indices of larval/juvenile growth with recruitment indices from the sablefish stock assessment except for 1997. Peterson et al. (2002) observed that oceanographic conditions were "normal" in early 1997. Surface temperatures during the time period examined in this study were low to moderate compared to other years. Indeed, the February PDO in 1997 was moderate and North Ekman transport in March was low within the time series, suggesting favorable conditions for early growth. Although upwelling began in April, conditions shifted abruptly starting in May and intense El Niño conditions persisted throughout the summer and winter of 1997 and into the spring and early summer of 1998. In waters off Oregon where juvenile sablefish reside, the shift resulted in downwelling, with poleward rather than equatorward transport, anomalously warm temperatures, reduced nitrate concentrations, and a shift in the zooplankton community with decreased copepod biomass and dominance by subtropical rather than boreal species (Peterson et al. 2002). The lowest growth index for the time series of sablefish examined in the present study occurred for the 1998 year class. El Niño conditions were particularly intense during the time period when this year class was incubating, undergoing embryo development, migrating to the surface, and actively foraging in the neuston layer. The physical conditions associated with the El Niño ended in late summer 1998, with a shift to an intense La Niña environment. Indeed, conditions reverted so extensively that 1999 was determined to begin a regime shift (Peterson and Schwing 2003), with conditions in the northern California Current matching the negative phase of the PDO. Thus, foraging conditions for larval and juvenile sablefish should have dramatically improved from 1998 to 1999, and this was clearly evident

in the growth index as well as the recruitment index in 1999, which was the highest of the time series. Poor recruitment of the 1997 year class was potentially a consequence of the oceanographic conditions subsequent to the early neustonic period, after the full El Niño environment developed.

CONCLUSIONS

In summary, growth of late larval and early juvenile stage sablefish off Oregon was rapid but variable across an 8-yr time series, with rates approaching those previously observed in lab studies. Growth rates increased markedly with increased fish size and warmer temperatures. Interannual differences corresponded to broad scale oceanographic conditions, with faster growth during years with presumed higher overall productivity based on indices such as the PDO. Recruitment estimates from stock assessments for the West Coast population of sablefish were correlated with early growth except in 1997, when growth was high but recruitment was extremely low. Because oceanographic conditions were generally favorable in early 1997, during the period of growth analysis for this study, but poor later in 1997, when intense El Niño conditions developed, poor recruitment was potentially related to poor survival after the early neustonic stage. These results are consistent with longer-term observations of strong environmental drivers of recruitment in West Coast sablefish (Schirripa and Colbert 2006) and suggest that bottom-up processes during the early life stages are important determinants of year class success.

ACKNOWLEDGMENTS

I am most grateful to N Kashef, D Stafford, and N Parker for assistance with otolith preparation and analysis. Numerous volunteers assisted with collections of sablefish. I thank Captains T Thompson (FV OLYMPIC) and B Eder (FV MICHELE ANN) for their expertise and enthusiasm during late night cruises. M Spencer, E Sturm, and M Ottmar provided support with laboratory processing of collections. J Field and two anonymous reviewers provided valuable comments on an earlier draft of the manuscript.

LITERATURE CITED

- Alderdice DF, Jensen JOT, Velsen FPJ. 1988. Preliminary trials on incubation of sablefish eggs (*Anoplopoma fimbria*). *Aquacult Fish Manage.* 69:271–290.
- Arendt JD. 1997. Adaptive intrinsic growth rates: an integration across taxa. *Quart Rev Biol.* 72:149–177. <http://dx.doi.org/10.1086/419764>
- Beamish RJ, McFarlane GA. 2000. Revaluation of the interpretation of annuli from otoliths of a long lived fish, *Anoplopoma fimbria*. *Fish Res.* 46:105–111. [http://dx.doi.org/10.1016/S0165-7836\(00\)00137-5](http://dx.doi.org/10.1016/S0165-7836(00)00137-5)
- Boehlert GW, Yoklavich MM. 1985. Larval and juvenile growth of sablefish, *Anoplopoma fimbria*, as determined from otolith increments. *Fish Bull.* 83:475–481.
- Brodeur RD, Peterson WT, Auth TD, Soulen HL, Parnel MM, Emerson AA. 2008. Abundance and diversity of coastal fish larvae as indicators of recent changes in ocean and climate conditions in the Oregon upwelling zone. *Mar Ecol Prog Ser.* 366:187–202. <http://dx.doi.org/10.3354/meps07539>
- Campana SE. 1990. How reliable are growth back-calculations based on otoliths? *Can J Fish Aquat Sci.* 47:2219–2227. <http://dx.doi.org/10.1139/f90-246>

- Francis RC, Hare SR, Hollowed AB, Wooster WS. 1998. Effects of interdecadal climate variability on the oceanic ecosystems of the NE Pacific. *Fish Oceanogr.* 7:1–21. <http://dx.doi.org/10.1046/j.1365-2419.1998.00052.x>
- Grover JJ, Olla BL. 1987. Effects of an El Niño event on the food habits of larval sablefish, *Anoplopoma fimbria*, off Oregon and Washington. *Fish Bull.* 85:71–79.
- Hare JA, Cowen RK. 1995. Effect of age, growth rate, and ontogeny on the otolith size—fish size relationship in bluefish, *Pomatomus saltatrix*, and the implications for back-calculation of size in fish early life history stages. *Can J Fish Aquat Sci.* 52:1909–1922. <http://dx.doi.org/10.1139/f95-783>
- Hooff RC, Peterson WT. 2006. Copepod biodiversity as an indicator of changes in ocean and climate conditions of the northern California current ecosystem. *Limnol Oceanogr.* 51:2607–2620. <http://dx.doi.org/10.4319/lo.2006.51.6.2607>
- Houde ED. 1987. Fish early life dynamics and recruitment variability. *Amer Fish Soc Symp.* 2:17–29.
- Johnson DW, Hixon MA. 2010. Ontogenetic and spatial variation in size-selective mortality of a marine fish. *J Evol Biol.* 23:724–737. PMID:20149026. <http://dx.doi.org/10.1111/j.1420-9101.2010.01938.x>
- Kendall Jr. AW, Matarese AC. 1987. Biology of eggs, larvae, and epipelagic juveniles of sablefish, *Anoplopoma fimbria*, in relation to their potential use in management. *Mar Fish Rev.* 49:1–13.
- Kimura DK, Shimada AM, Shaw FR. 1998. Stock structure and movement of tagged sablefish, *Anoplopoma fimbria*, in offshore northeast Pacific water and the effects of El Niño–Southern Oscillation on migration and growth. *Fish Bull.* 96:462–481.
- King JR, McFarlane GA, Beamish RJ. 2000. Decadal-scale patterns in the relative year class success of sablefish (*Anoplopoma fimbria*). *Fish Oceanogr.* 9:62–70. <http://dx.doi.org/10.1046/j.1365-2419.2000.00122.x>
- King JR, McFarlane GA, Beamish RJ. 2001. Incorporating the dynamics of marine systems into the stock assessment and management of sablefish. *Prog Oceanogr.* 49:619–639. [http://dx.doi.org/10.1016/S0079-6611\(01\)00044-1](http://dx.doi.org/10.1016/S0079-6611(01)00044-1)
- Lee RF, Hagen W, Kattner G. 2006. Lipid storage in marine zooplankton. *Mar Ecol Prog Ser.* 307:273–306. <http://dx.doi.org/10.3354/meps307273>
- Lehodey P, Alheit J, Barange M, Baumgartner T, Beaugrand G, Drinkwater K, Fromentin J, Hare SR, Ottersen G, Perry RI, et al. 2006. Climate variability, fish, and fisheries. *J Climate.* 19:5009–5030. <http://dx.doi.org/10.1175/JCLI3898.1>
- Mantua NJ, Hare SR, Zhang Y, Wallace SM, Francis RC. 1997. A Pacific interdecadal climate oscillation with impacts on salmon production. *Bull Amer Meteor Soc.* 78:1069–1080. [http://dx.doi.org/10.1175/1520-0477\(1997\)078<1069:APICOW>2.0.CO;2](http://dx.doi.org/10.1175/1520-0477(1997)078<1069:APICOW>2.0.CO;2)
- Mason JC, Beamish RJ, McFarlane GA. 1983. Sexual maturity, fecundity, spawning, and early life history of sablefish (*Anoplopoma fimbria*) off the Pacific Coast of Canada. *Can J Fish Aquat Sci.* 40:2126–2134. <http://dx.doi.org/10.1139/f83-247>
- McFarlane GA, Beamish RJ. 1992. Climatic influence linking copepod production with strong year-classes in sablefish, *Anoplopoma fimbria*. *Can J Fish Aquat Sci.* 49:743–753. <http://dx.doi.org/10.1139/f92-083>
- Parrish RH, Nelson CS, Bakun A. 1981. Transport mechanisms and reproductive success of fishes in the California Current. *Biol Oceanogr.* 1:175–203.
- Peterson WT, Keister JE, Feinberg LR. 2002. The effects of the 1997–99 El Niño/La Niña events on hydrography and zooplankton off the central Oregon coast. *Prog Oceanogr.* 54:381–398. [http://dx.doi.org/10.1016/S0079-6611\(02\)00059-9](http://dx.doi.org/10.1016/S0079-6611(02)00059-9)
- Peterson WT, Schwing FB. 2003. A new climate regime in northeast Pacific ecosystems. *Geophys Res Lett.* 30.
- Schirripa MJ, Colbert JJ. 2006. Interannual changes in sablefish (*Anoplopoma fimbria*) recruitment in relation to oceanographic conditions within the California Current System. *Fish Oceanogr.* 15:25–36. <http://dx.doi.org/10.1111/j.1365-2419.2005.00352.x>

- Schroeder ID, Sydeman WJ, Sarkar N, Thompson SA, Bograd SJ, Schwing FB. 2009. Winter pre-conditioning of seabird phenology in the California Current. *Mar Ecol Prog Ser.* 393:211–223. <http://dx.doi.org/10.3354/meps08103>
- Shanks AL, Eckert GL. 2005. Population persistence of California Current fishes and benthic crustaceans: a marine drift paradox. *Ecol Monogr.* 75:505–524. <http://dx.doi.org/10.1890/05-0309>
- Shenker JM. 1988. Oceanographic associations of neustonic larval and juvenile fishes and dungeness crab megalopae off Oregon. *Fish Bull.* 86:299–317.
- Sigler MF, Rutecki TL, Courtney OL, Karinen JF, Yang M-S. 2001. Young of the year sablefish abundance, growth, and diet in the Gulf of Alaska. *Alaska Fish Res Bull.* 8:57–70.
- Sogard SM, Olla BL. 1998. Behavior of juvenile sablefish, *Anoplopoma fimbria* (Pallas), in a thermal gradient: balancing food and temperature requirements. *J Exp Mar Biol Ecol.* 222:43–58. [http://dx.doi.org/10.1016/S0022-0981\(97\)00137-8](http://dx.doi.org/10.1016/S0022-0981(97)00137-8)
- Sogard SM, Olla BL. 2001. Growth and behavioral responses to elevated temperatures by juvenile sablefish *Anoplopoma fimbria* and the interactive role of food availability. *Mar Ecol Prog Ser.* 217:121–134. <http://dx.doi.org/10.3354/meps217121>
- Sogard SM, Olla BL. 2002. Contrasts in the capacity and underlying mechanisms for compensatory growth in two pelagic marine fishes. *Mar Ecol Prog Ser.* 243:165–177. <http://dx.doi.org/10.3354/meps243165>
- Sogard SM, Spencer ML. 2004. Energy allocation in juvenile sablefish: effects of temperature, ration and body size. *J Fish Biol.* 64:726–738. <http://dx.doi.org/10.1111/j.1095-8649.2004.00342.x>
- Wells BK, Field JC, Thayer JA, Grimes CB, Bograd SJ, Sydeman WJ, Schwing FB, Hewitt R. 2008. Untangling the relationships among climate, prey and top predators in an ocean ecosystem. *Mar Ecol Prog Ser.* 364:15–29. <http://dx.doi.org/10.3354/meps07486>
- Wexler JB, Chow S, Wakabayashi T, Nohara K, Margulies D. 2007. Temporal variation in growth of yellowfin tuna (*Thunnus albacares*) larvae in the Panama Bight, 1990–97. *Fish Bull.* 105:1–18.

DATE SUBMITTED: 21 May, 2010.

DATE ACCEPTED: 31 August, 2010.

AVAILABLE ONLINE: 21 September, 2010.

ADDRESS: Fisheries Ecology Division, Southwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, 110 Shaffer Road, Santa Cruz, California 95060. E-mail: <susan.sogard@noaa.gov>.

